

SightOS — The AI Operating System for Visual Health Intelligence

“Every human deserves perfect vision. AI makes it possible.”

Executive Summary

SightOS is the comprehensive AI platform transforming visual health from reactive treatment to proactive optimization. By integrating continuous monitoring, early disease detection, personalized interventions, and smart device orchestration, SightOS creates a unified intelligence layer for the \$180B global eye care market.

The Billion-Dollar Thesis: Vision is humanity’s most critical sense, yet eye care remains fragmented, reactive, and inaccessible. 2.2 billion people globally have vision impairment. 50% of vision loss is preventable. SightOS becomes the operating system connecting patients, providers, devices, and AI to preserve and optimize vision at population scale.

The Problem

The Vision Crisis Is Exploding

The Numbers Are Staggering: - **2.2 billion people** globally have vision impairment - **1 billion** cases are preventable or untreated - **89%** of visually impaired people live in low/middle-income countries - **\$411B** annual global economic impact of vision impairment - **Myopia epidemic:** 50% of world population projected myopic by 2050

Screen Time Is Destroying Eyes: - Average adult: 7+ hours daily screen time - Children: 4-6 hours daily screen exposure - Digital eye strain affects 65% of Americans - Blue light exposure linked to retinal damage - Near-work myopia exploding in youth

Eye Care Is Broken: - Average 2-year gap between eye exams - Diabetic retinopathy: 50% undiagnosed until vision loss - Glaucoma: 50% unaware they have it - Macular degeneration: Often caught too late - Rural areas: Severe optometrist shortages

The Market Gap

No platform exists that: - Continuously monitors visual health between exams - Detects disease progression in real-time - Personalizes interventions based on lifestyle - Coordinates care across providers and devices - Makes professional eye care accessible anywhere

The Solution

SightOS: Your Eyes’ Operating System

SightOS creates an AI-powered visual health ecosystem:

SIGHTOS PLATFORM

Vision	Disease	Behavior	Device
Tracking	Detection	Coaching	Orchestra

SIGHTOS AI ENGINE

- Computer Vision Analysis • Progression Models
- Lifestyle Integration • Treatment Optimization

Smartphones & Cameras	Smart Glasses	Wearables & Sensors	Clinical Equipment
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Core Capabilities

- 1. Continuous Vision Monitoring** - Smartphone-based visual acuity tracking - Contrast sensitivity measurements - Color vision assessment - Peripheral vision mapping - Eye movement analysis - Pupil response monitoring
- 2. AI Disease Detection** - Retinal imaging analysis via smartphone - Diabetic retinopathy screening - Glaucoma risk assessment - Macular degeneration detection - Cataracts progression tracking - Keratoconus early warning
- 3. Behavioral Eye Health** - Screen time impact analysis - Blue light exposure tracking - Blink rate monitoring - Eye strain detection - Posture and distance coaching - 20-20-20 rule automation
- 4. Personalized Interventions** - Adaptive display settings - Exercise programs for eye muscles - Nutrition recommendations - Sleep optimization for eye health - Environmental adjustments - Lens prescription predictions
- 5. Smart Device Orchestration** - Smart glasses integration - Auto-adjusting monitor settings - Ambient lighting control - E-reader optimization - VR/AR eye safety - Prescription update predictions

Product Architecture

For Consumers

SightOS Personal — \$9.99/month - Daily vision check-ins (30 seconds) - Screen time analytics and coaching - Blue light management - Eye strain alerts and breaks - Basic disease risk screening - Personalized eye exercises

SightOS Family — \$19.99/month - Up to 6 family members - Pediatric myopia prevention - Age-related monitoring for parents - Family visual health dashboard - Shared care coordination - School vision requirements tracking

SightOS Premium — \$29.99/month - Advanced AI disease screening - Telehealth optometrist access - Prescription management - Smart device integrations - Genetic risk analysis - Priority support

For Providers

SightOS Clinic — Per-provider pricing - AI-assisted diagnostics - Patient monitoring between visits - Automated risk stratification - Treatment outcome tracking - Population health analytics - EMR integrations

SightOS Enterprise — Custom - Large practice management - Multi-location analytics - Clinical trial recruitment - Referral network optimization - Revenue cycle enhancement

For Employers

SightOS Workplace — Per-employee pricing - Employee eye health programs - Ergonomic display optimization - Screen fatigue reduction - Workers' comp prevention - Productivity analytics - Vision benefits

Technology Deep Dive

AI Vision Analysis Engine

Smartphone Fundoscopy:

```
class RetinalAnalyzer:
    """
    Analyzes retinal images from smartphone cameras
    with ophthalmologist-level accuracy
    """

    def analyze_fundus_image(self, image):
        # Preprocessing for smartphone quality
        enhanced = self.enhance_image(image)

        # Multi-disease screening
        results = {
            'diabetic_retinopathy': self.dr_model.predict(enhanced),
            'glaucoma_risk': self.glaucoma_model.predict(enhanced),
            'amd_signs': self.amd_model.predict(enhanced),
            'vessel_analysis': self.vessel_model.analyze(enhanced),
            'optic_disc': self.disc_model.assess(enhanced)
        }

        # Generate clinical report
        return self.generate_report(results)

    def track_progression(self, patient_id, new_scan):
        # Compare against historical scans
        history = self.get_patient_history(patient_id)
        delta = self.compute_progression(history, new_scan)

        if delta.requires_attention:
            self.alert_care_team(patient_id, delta)

        return delta
```

Visual Function Tracking:

```
class VisionTracker:
    """
    Continuous visual function monitoring through
    daily micro-assessments
    """

    def daily_check(self, user):
        assessments = []

        # Quick visual acuity (Landolt C test)
        acuity = self.measure_acuity(user)
        assessments.append(acuity)
```

```

# Contrast sensitivity
contrast = self.measure_contrast(user)
assessments.append(contrast)

# Color vision spot check
if user.schedule.includes_color_today:
    color = self.measure_color_vision(user)
    assessments.append(color)

# Eye movement tracking
movements = self.analyze_eye_movements(user)
assessments.append(movements)

# Aggregate and detect changes
return self.analyze_trends(user, assessments)

```

Behavioral Intelligence

Screen Time Impact Model: - Correlates screen habits with visual changes - Predicts eye strain before it occurs - Optimizes break timing for each user - Adjusts display settings dynamically - Integrates with device ecosystems

Myopia Prevention Engine: - Outdoor time tracking and encouragement - Near-work duration alerts - Reading distance monitoring - Ambient light optimization - Atropine/ortho-k coordination

Device Integration Layer

Universal Protocol: - Works with any smart display - Integrates smart glasses (Apple, Meta, etc.) - Controls ambient lighting systems - Syncs with fitness wearables - Coordinates with hearing devices

Go-to-Market Strategy

Phase 1: Consumer Foundation (Months 1-12)

Target Segments: 1. **Digital Workers** — Remote workers, programmers, gamers 2. **Parents of Young Children** — Myopia prevention 3. **Diabetics** — Retinopathy screening 4. **Adults 40+** — Age-related monitoring

Launch Strategy: - Free basic vision tracker app - Viral “Know Your Eye Age” assessment - Influencer partnerships (eye health awareness) - App Store optimization - Content marketing (blue light truth, screen health)

Metrics: - 1M downloads in 6 months - 100K paid subscribers by month 12 - 4.5+ app store rating - <5% monthly churn

Phase 2: Healthcare Integration (Months 6-18)

Provider Onboarding: - Partner with 100 progressive optometry practices - EMR integrations (Epic, Cerner, etc.) - Clinical validation studies - Continuing education programs - Referral network establishment

Payer Relationships: - Vision insurance partnerships (VSP, EyeMed) - Medical insurance pilots (diabetic screening) - Medicare Advantage programs - Employer vision benefits

Phase 3: Enterprise & Global (Months 12-24)

Workplace Wellness: - Fortune 500 employee programs - Tech company partnerships - Manufacturing/industrial applications - Government/military contracts

International Expansion: - India: Mass screening programs - China: Myopia epidemic intervention - Europe: GDPR-compliant deployment - Africa: Mobile-first screening

Business Model

Revenue Streams

1. Consumer Subscriptions | Tier | Price | Target Users | Revenue Potential | |——|——|———|———|———|———|
| Personal | \$9.99/mo | 2M users | \$240M ARR | | Family | \$19.99/mo | 500K families | \$120M ARR | | Premium | \$29.99/mo | 300K users | \$108M ARR |

2. Healthcare SaaS | Product | Price | Target | Revenue Potential | |——|——|———|———|———|———|
| Clinic | \$500/provider/mo | 10K providers | \$60M ARR | | Enterprise | \$100K+/year | 200 systems | \$20M ARR |

3. Employer Programs - Per-employee pricing: \$3-8/employee/month - Target: 5M covered employees - Revenue potential: \$180M-480M ARR

4. Licensing & Partnerships - Device manufacturer licensing - Pharma partnership revenue - Insurance data partnerships - Research collaborations

Year 5 Revenue Target: \$500M-800M ARR

Unit Economics

Consumer: - CAC: \$25 (organic + paid) - LTV: \$180 (18-month average tenure) - LTV/CAC: 7.2x

Enterprise: - CAC: \$15,000 - ACV: \$120,000 - LTV: \$400,000 - LTV/CAC: 26x

Competitive Landscape

Current Players

Company	Focus	Limitation
EyeQue	At-home refraction	No disease detection
Eyenuk	Diabetic retinopathy	Clinical only, no consumer
Warby Parker	Retail + basic tracking	Surface-level monitoring
Smart glasses makers	Display, not health	Vision as afterthought

SightOS Differentiation

- 1. Comprehensive vs. Point Solution** — Full visual health, not just one metric
 - 2. Continuous vs. Episodic** — Daily monitoring, not annual exams
 - 3. Consumer + Clinical** — Bridges home and professional care
 - 4. Platform vs. Device** — Works with any hardware
 - 5. AI-Native** — Built for machine learning from day one
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Regulatory Strategy

FDA Pathway

510(k) Clearance: - Visual acuity tracking: Class I exempt - Disease screening: Class II 510(k) - Predicate devices: IDx-DR, EyeArt

Clinical Validation: - Partner with academic medical centers - 10,000 patient validation studies - Comparative accuracy studies - Real-world evidence collection

Timeline: - Month 6: IDE submission for disease detection - Month 12: 510(k) submission - Month 18: Clearance expected

International

- CE Mark (Europe): Parallel filing
 - Health Canada: Class II pathway
 - TGA (Australia): Expedited review eligible
 - PMDA (Japan): Partner with local distributor
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Team Requirements

Founding Team

CEO: Consumer health executive with scale experience **CTO:** Computer vision PhD with mobile expertise **CMO:** Ophthalmologist with digital health background **CPO:** Consumer product leader (health/wellness apps) **VP Engineering:** Mobile platform architect **VP Regulatory:** FDA/medical device experience

Key Hires (Year 1)

- AI/ML team (5-8 engineers)
 - Clinical affairs team (3-4)
 - Mobile engineering (6-8)
 - Product/design (4-5)
 - Go-to-market (5-6)
 - Customer success (3-4)
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Financial Projections

5-Year Model

Metric	Year 1	Year 2	Year 3	Year 4	Year 5
Consumer Users	100K	500K	1.5M	3M	5M
Enterprise Clients	50	200	500	1,000	2,000
ARR	\$15M	\$60M	\$180M	\$400M	\$750M
Gross Margin	75%	78%	80%	82%	83%
Employees	50	120	250	400	600

Funding Requirements

Seed Round: \$5M - MVP development - Clinical partnerships - Initial user acquisition

Series A: \$25M - Product expansion - FDA clearance - Market launch

Series B: \$75M - Scale go-to-market - International expansion - Platform buildout

Series C: \$150M - Category dominance - Strategic acquisitions - Enterprise acceleration

Risk Mitigation

Risk	Mitigation
Regulatory delays	Parallel submission strategy, predicate selection
Clinical accuracy	Extensive validation, conservative claims
Consumer adoption	Freemium model, viral mechanics
Competition	Speed, comprehensive platform, data moat
Data privacy	SOC 2, HIPAA, GDPR from day one
Reimbursement	Value-based care partnerships, employer direct

Exit Scenarios

Strategic Acquirers

1. **Apple/Google** — Health platform integration (\$3-5B)
2. **Meta** — Smart glasses ecosystem (\$2-4B)
3. **Johnson & Johnson Vision** — Digital health expansion (\$2-3B)
4. **VSP/EyeMed** — Vertical integration (\$1.5-2.5B)
5. **Teladoc/Amwell** — Telehealth expansion (\$1-2B)

IPO Path

- Year 5: \$500M+ ARR
 - 30%+ growth rate
 - Clear market leadership
 - Public market ready
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Why Now?

Convergent Trends

1. **Smartphone Cameras** — Now capable of medical-grade imaging
 2. **AI Maturity** — Vision models achieve clinical accuracy
 3. **Screen Epidemic** — Digital eye strain is universal pain
 4. **Aging Population** — Age-related eye disease surging
 5. **Telehealth Acceptance** — Remote monitoring normalized
 6. **Smart Glasses Era** — Vision-first devices emerging
 7. **Preventive Health** — Shift from treatment to prevention
 8. **Value-Based Care** — Payers reward early detection
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The Moonshot

Imagine a world where: - No one goes blind from preventable causes - Every child's myopia is caught and slowed - Diabetics never lose vision to retinopathy - Screen time doesn't destroy our eyes - Perfect vision

is optimized, not just corrected - Eye exams happen daily, not yearly - The 1 billion with untreated vision impairment get care

SightOS makes this world possible.

Vision is humanity's most precious sense. It's time we had an operating system to protect it.

"The eye is the window to the soul — and the body. SightOS keeps that window clear."

Appendix

Market Size Details

Global Eye Care Market: \$180B (2026) - Prescription eyewear: \$90B - Eye care services: \$50B - Ophthalmic drugs: \$25B - Surgical equipment: \$15B

Digital Eye Health Market: \$8B (2026) - Growing 15% annually - SightOS TAM: \$30B by 2030

Clinical Evidence

AI Diagnostic Accuracy: - Diabetic retinopathy: 96% sensitivity (vs. 82% general ophthalmologists) - Glaucoma detection: 94% AUC - AMD staging: 91% accuracy

Smartphone Imaging: - Studies show smartphone fundoscopy achieves clinical quality - \$20 adapter enables retinal imaging - Algorithm enhancement compensates for hardware limitations

Technical Requirements

User Device: - iOS 14+ / Android 10+ - Camera: 12MP+ main sensor - Display: 60Hz+ refresh rate - Connectivity: 4G/WiFi

Backend: - HIPAA-compliant cloud (AWS/GCP) - Edge inference for latency-sensitive features - Federated learning for privacy-preserving model updates

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This document is confidential and intended for potential investors and partners.